

Chapter 1

Introduction, vectors and basic calculus

The nature of physics

- Physics is an *experimental* science in which physicists seek patterns that relate the phenomena of nature.
- The patterns are called **physical theories**.
- A very well established or widely used theory is called a **physical law** or **principle**.

According to legend, Galileo investigated falling objects by dropping them from the Leaning Tower of Pisa, Italy, ...



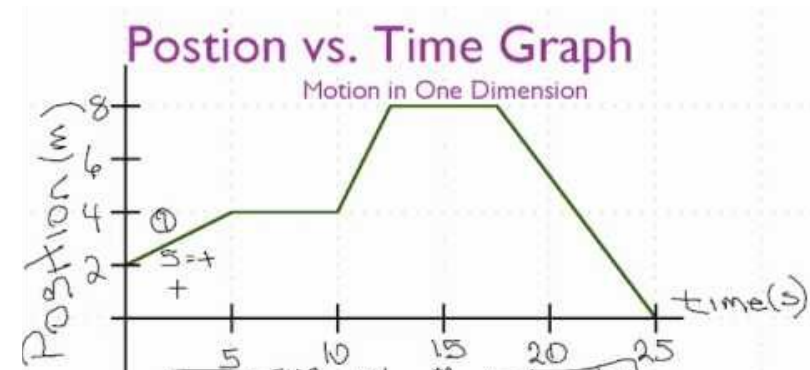
... and he studied pendulum motion by observing the swinging chandelier in the adjacent cathedral.

What is physics?

- For example, the description of motion
- You use a ruler to **measure** the position and a stopwatch to measure the time
- Plot a position and time graph to **describe** the motion
- The motion can be **explained** using **Newton's law**

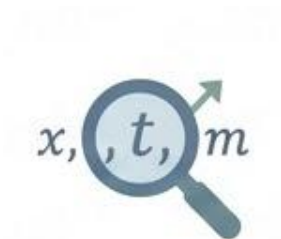
Newton's law

$$\vec{F} = m\vec{a}$$



Solving problems in physics

- All of the *Problem-Solving Strategies* and *Examples* in this book will follow these four steps:
- **Identify** the relevant concepts, target variables, and known quantities, as stated or implied in the problem.
- **Set Up** the problem: Choose the equations that you'll use to solve the problem, and draw a sketch of the situation.
- **Execute** the solution: This is where you “do the math.”
- **Evaluate** your answer: Compare your answer with your estimates, and reconsider things if there's a discrepancy.



Idealized models

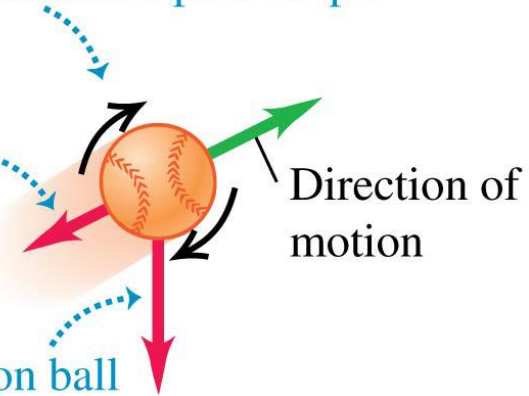
To simplify the analysis of (a) a baseball in flight, we use (b) an idealized model.

(a) A real baseball in flight

Baseball spins and has a complex shape.

Air resistance and wind exert forces on the ball.

Gravitational force on ball depends on altitude.

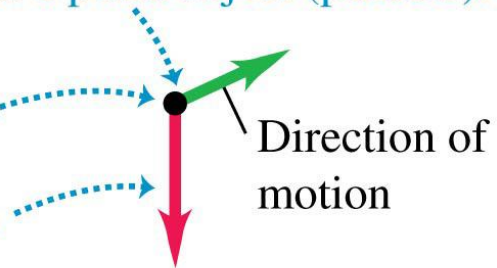


(b) An idealized model of the baseball

Treat the baseball as a point object (particle).

No air resistance.

Gravitational force on ball is constant.



Standards and units

- Length, time, and mass are three *fundamental* quantities of physics.
- The *International System* (SI for *Système International*) is the most widely used system of units.
- In SI units, length is measured in *meters*, time in *seconds*, and mass in *kilograms*.

Unit prefixes

- Prefixes can be used to create larger and smaller units for the fundamental quantities. Some examples are:
- $1\ \mu\text{m} = 10^{-6}\ \text{m}$ (size of some bacteria and living cells)
- $1\ \text{km} = 10^3\ \text{m}$ (a 10-minute walk)
- $1\ \text{mg} = 10^{-6}\ \text{kg}$ (mass of a grain of salt)
- $1\ \text{g} = 10^{-3}\ \text{kg}$ (mass of a paper clip)
- $1\ \text{ns} = 10^{-9}\ \text{s}$ (time for light to travel 0.3 m)

Unit prefixes

Name	Symbol	Value
kilo	k	10^3
mega	M	10^6
giga	G	10^9
tera	T	10^{12}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}
femto	f	10^{-15}

Greek letters

$\text{A } \alpha$ Alpha	$\text{B } \beta$ Beta	$\text{Γ } \gamma$ Gamma	$\text{Δ } \delta$ Delta	$\text{E } \varepsilon$ Epsilon	$\text{Z } \zeta$ Zeta
$\text{H } \eta$ Eta	$\text{Θ } \theta$ Theta	$\text{I } \iota$ Iota	$\text{K } \kappa$ Kappa	$\text{Λ } \lambda$ Lambda	$\text{M } \mu$ Mu
$\text{N } \nu$ Nu	$\text{Ξ } \xi$ Xi	$\text{O } \omicron$ Omicron	$\text{Π } \pi$ Pi	$\text{P } \rho$ Rho	$\text{Σ } \sigma, \varsigma$ Sigma
$\text{T } \tau$ Tau	$\text{Υ } \upsilon$ Upsilon	$\text{Φ } \phi$ Phi	$\text{X } \chi$ Chi	$\text{Ψ } \psi$ Psi	$\text{Ω } \omega$ Omega

Unit consistency and conversions

- An equation must be *dimensionally consistent*. Terms to be added or equated must *always* have the same units. (Be sure you're adding “apples to apples.”)
- Always carry units through calculations.
- Convert to standard units as necessary, by forming a ratio of the same physical quantity in two different units, and using it as a multiplier.
- For example, to find the number of seconds in 3 min, we write:

$$3 \text{ min} = (3 \text{ min}) \left(\frac{60 \text{ s}}{1 \text{ min}} \right) = 180 \text{ s}$$

Uncertainty and significant figures

- The uncertainty of a measured quantity is indicated by its number of *significant figures*.
- For multiplication and division, the answer can have no more significant figures than the *smallest* number of significant figures in the factors.
- For addition and subtraction, the number of significant figures is determined by the term having the fewest digits to the right of the decimal point.
- As this train mishap illustrates, even a small percent error can have spectacular results!



Montparnasse
derailment

Vectors and scalars

- A **scalar quantity** can be described by a *single number*.
- A **vector quantity** has both a *magnitude* and a *direction* in space.
- A vector quantity is typically represented by
 - an italic letter with an arrow over it: $\vec{A}$
 - or simply $\mathbf{A}$ in bold font when you type it.
- The magnitude of $\vec{A}$ is written as A (no bold font) or $|\vec{A}|$.
- A typical example of a vector: position of Kowloon Tong station relative to the Central station.

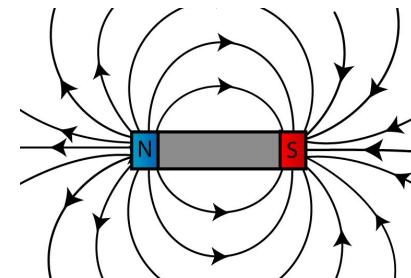
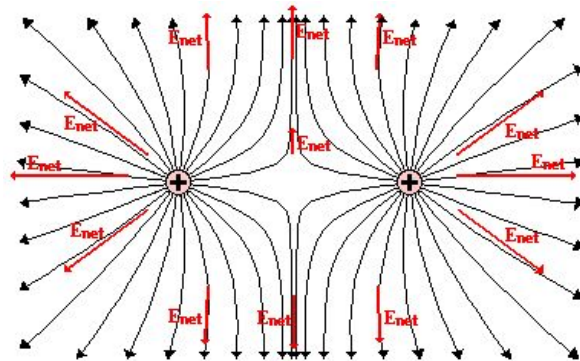
Vector (relative position)

- The relative position of Wu Kai Sha with respect to Kowloon Tong is a vector.
- It tells you how to get to Wu Kai Sha from Kowloon Tong.
- Walk a **distance** along a certain **direction**
- If the direction is changed, you go to the wrong place (red arrow)



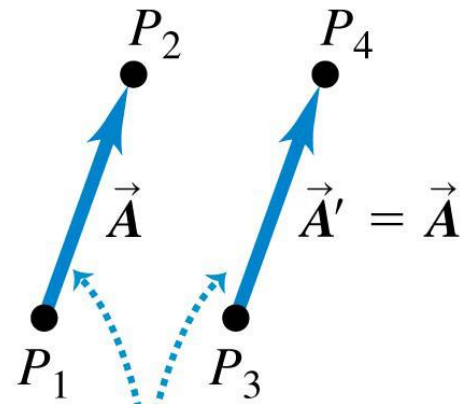
Other vector physical quantities

- Apart from displacement, the following physical quantities are also vectors
- Velocity, momentum (come from position)
- Angular velocity, angular momentum (come from position)
- Acceleration, force
- Electric field, magnetic field
- They are usually related to motion and force

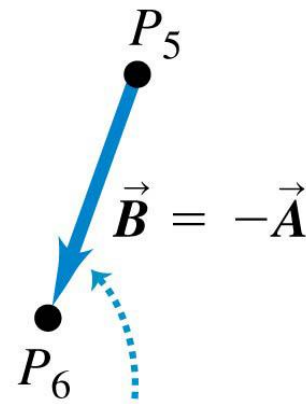


Drawing vectors

- Draw a vector as a line with an arrowhead at its tip.
- The *length* of the line shows the vector's *magnitude*.
- The *direction* of the line shows the vector's *direction*.



Displacements $\vec{A}$ and $\vec{A}'$ are equal because they have the same length and direction.



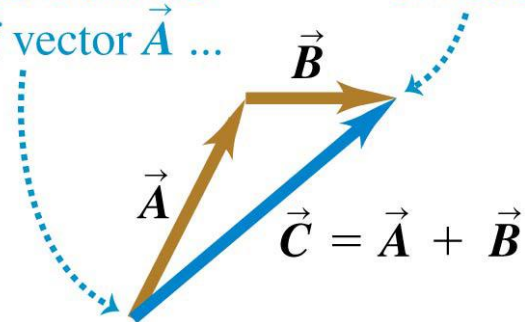
Displacement $\vec{B}$ has the same magnitude as $\vec{A}$ but opposite direction; $\vec{B}$ is the negative of $\vec{A}$.

Adding two vectors graphically

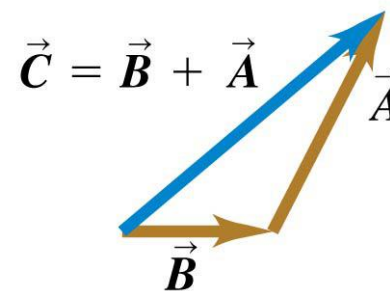
(a) We can add two vectors by placing them head to tail.

The vector sum $\vec{C}$ extends from the tail of vector $\vec{A}$...

... to the head of vector $\vec{B}$.

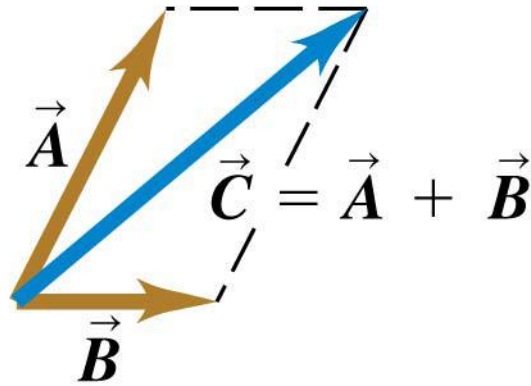


(b) Adding them in reverse order gives the same result: $\vec{A} + \vec{B} = \vec{B} + \vec{A}$. The order doesn't matter in vector addition.



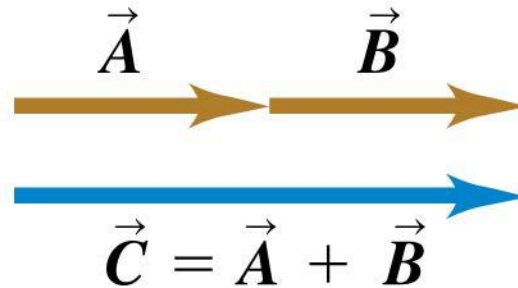
Adding two vectors graphically

(c) We can also add two vectors by placing them tail to tail and constructing a parallelogram.

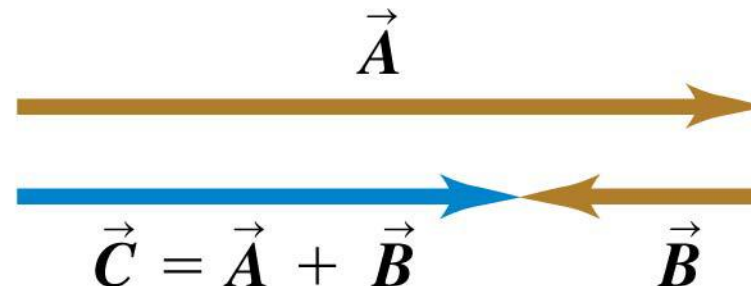


Adding two vectors graphically

(a) Only when vectors $\vec{A}$ and $\vec{B}$ are parallel does the magnitude of their vector sum $\vec{C}$ equal the sum of their magnitudes: $C = A + B$.



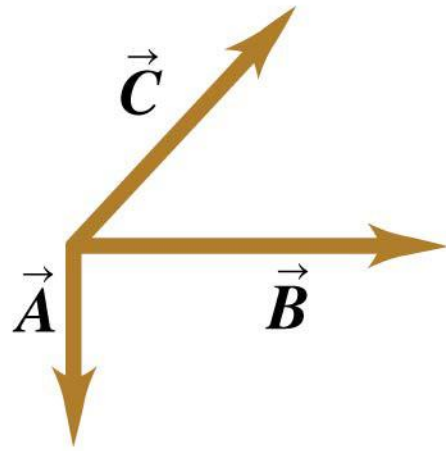
(b) When $\vec{A}$ and $\vec{B}$ are antiparallel, the magnitude of their vector sum $\vec{C}$ equals the *difference* of their magnitudes: $C = |A - B|$.



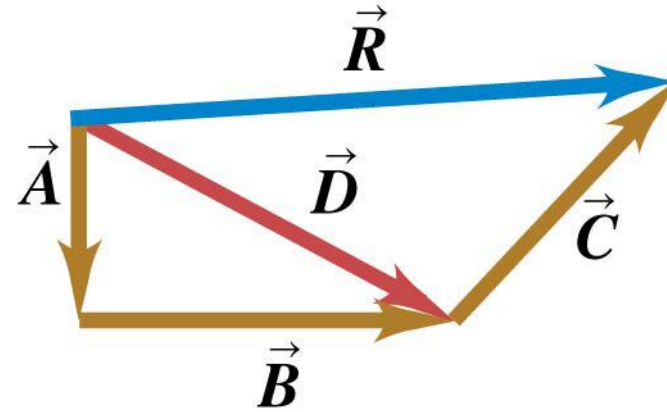
Adding more than two vectors graphically

- To add several vectors, use the head-to-tail method.
- The vectors can be added in any order.

(a) To find the sum of these three vectors ...



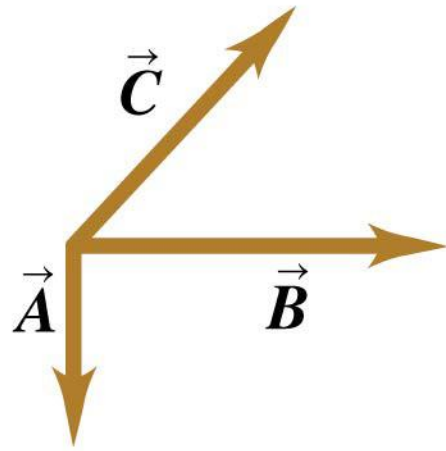
(b) ... add $\vec{A}$ and $\vec{B}$ to get $\vec{D}$ and then add $\vec{C}$ to $\vec{D}$ to get the final sum (resultant) $\vec{R}$...



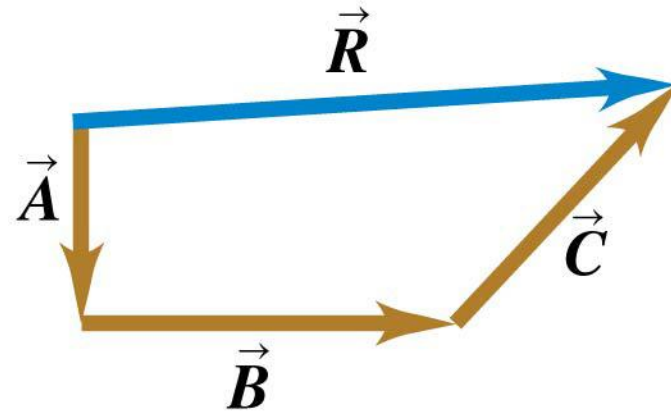
Adding more than two vectors graphically

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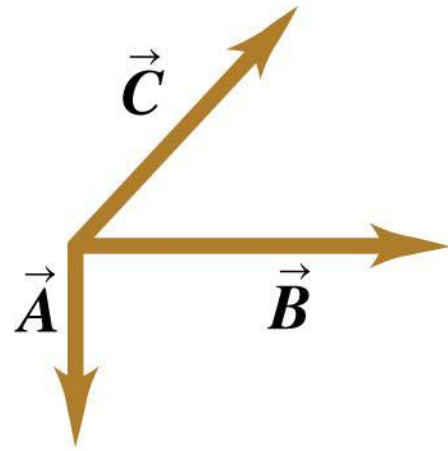
(d) ... or add $\vec{A}$, $\vec{B}$, and $\vec{C}$ to get $\vec{R}$ directly ...



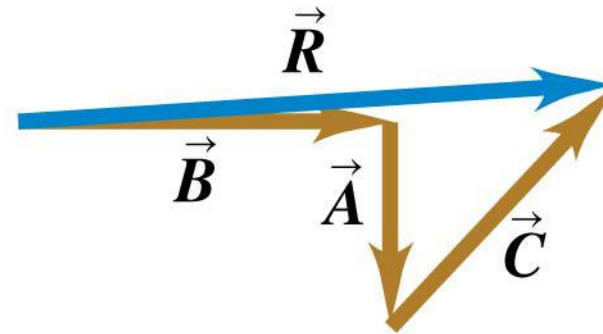
Adding more than two vectors graphically

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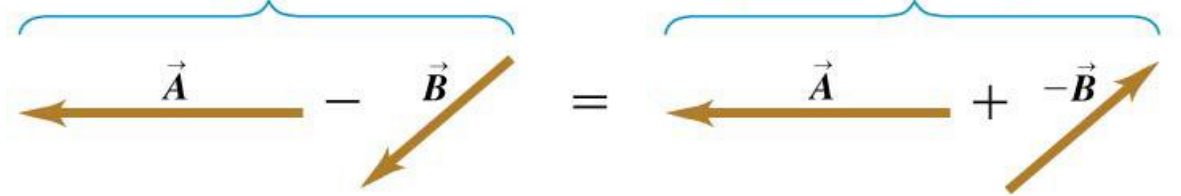
(e) ... or add $\vec{A}$, $\vec{B}$, and $\vec{C}$ in any other order and still get $\vec{R}$.



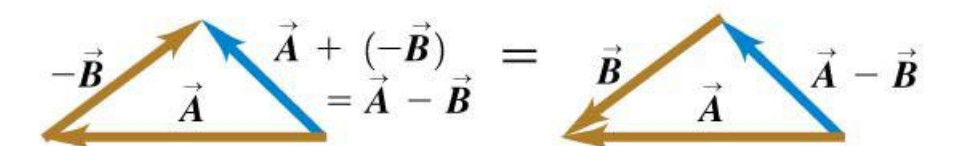
Subtracting vectors

Subtracting $\vec{B}$ from $\vec{A}$...

... is equivalent to adding $-\vec{B}$ to $\vec{A}$.



The diagram illustrates the equivalence between subtracting vector $\vec{B}$ from vector $\vec{A}$ and adding the negative vector $-\vec{B}$ to $\vec{A}$. On the left, vector $\vec{A}$ is represented by a horizontal arrow pointing left, and vector $\vec{B}$ is a diagonal arrow pointing down and to the right. A minus sign is between them. On the right, vector $\vec{A}$ is the same horizontal arrow pointing left, and vector $-\vec{B}$ is a diagonal arrow pointing up and to the right. A plus sign is between them. The two expressions are set equal to each other.

$$\vec{A} + (-\vec{B}) = \vec{A} - \vec{B}$$


The diagram shows two equivalent methods for finding the resultant vector $\vec{A} - \vec{B}$ using the triangle rule. In the left triangle, vector $\vec{A}$ is horizontal and points left, and vector $-\vec{B}$ is diagonal and points up and right. The resultant vector $\vec{A} - \vec{B}$ is the third side of the triangle, connecting the tail of $\vec{A}$ to the head of $-\vec{B}$. In the right triangle, vector $\vec{A}$ is horizontal and points left, and vector $\vec{B}$ is diagonal and points down and right. The resultant vector $\vec{A} - \vec{B}$ is the third side of the triangle, connecting the tail of $\vec{A}$ to the tail of $\vec{B}$.

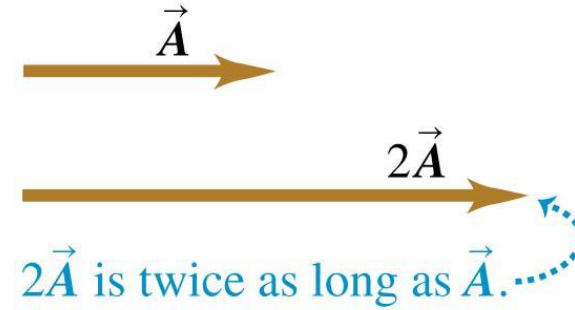
With $\vec{A}$ and $-\vec{B}$ head to tail, $\vec{A} - \vec{B}$ is the vector from the tail of $\vec{A}$ to the head of $-\vec{B}$.

With $\vec{A}$ and $\vec{B}$ head to head, $\vec{A} - \vec{B}$ is the vector from the tail of $\vec{A}$ to the tail of $\vec{B}$.

Multiplying a vector by a scalar

- If c is a scalar, the product $c\vec{A}$ has magnitude $|c|A$.
- The figure illustrates multiplication of a vector by (a) a positive scalar and (b) a negative scalar.

(a) Multiplying a vector by a positive scalar changes the magnitude (length) of the vector but not its direction.

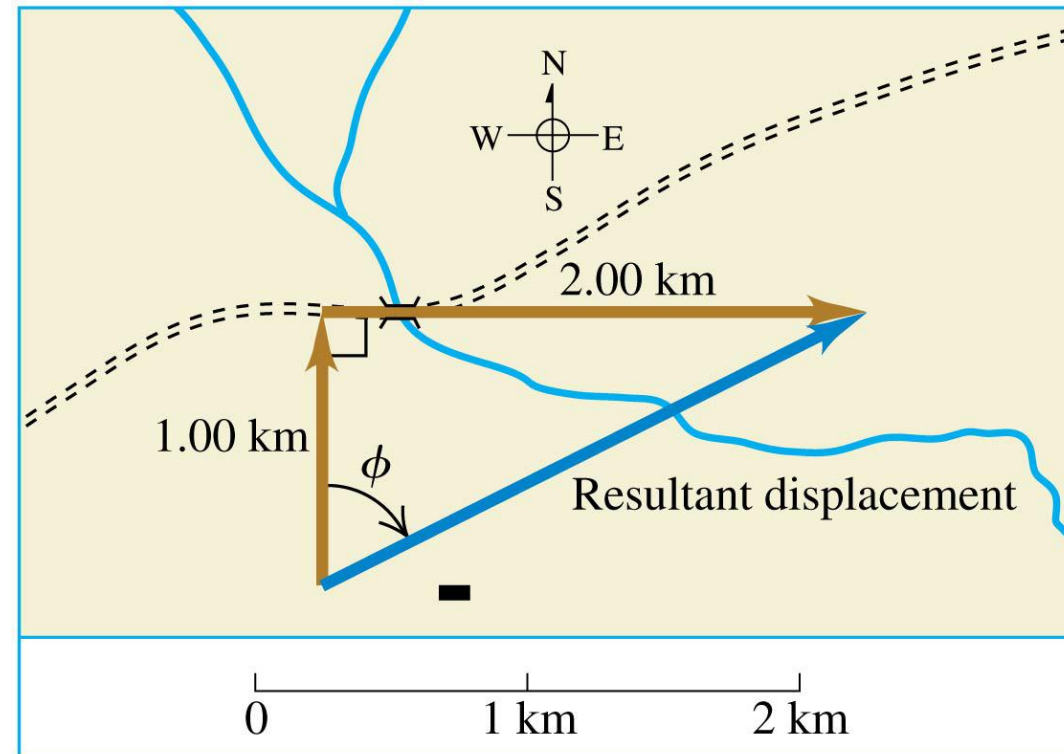


(b) Multiplying a vector by a negative scalar changes its magnitude and reverses its direction.



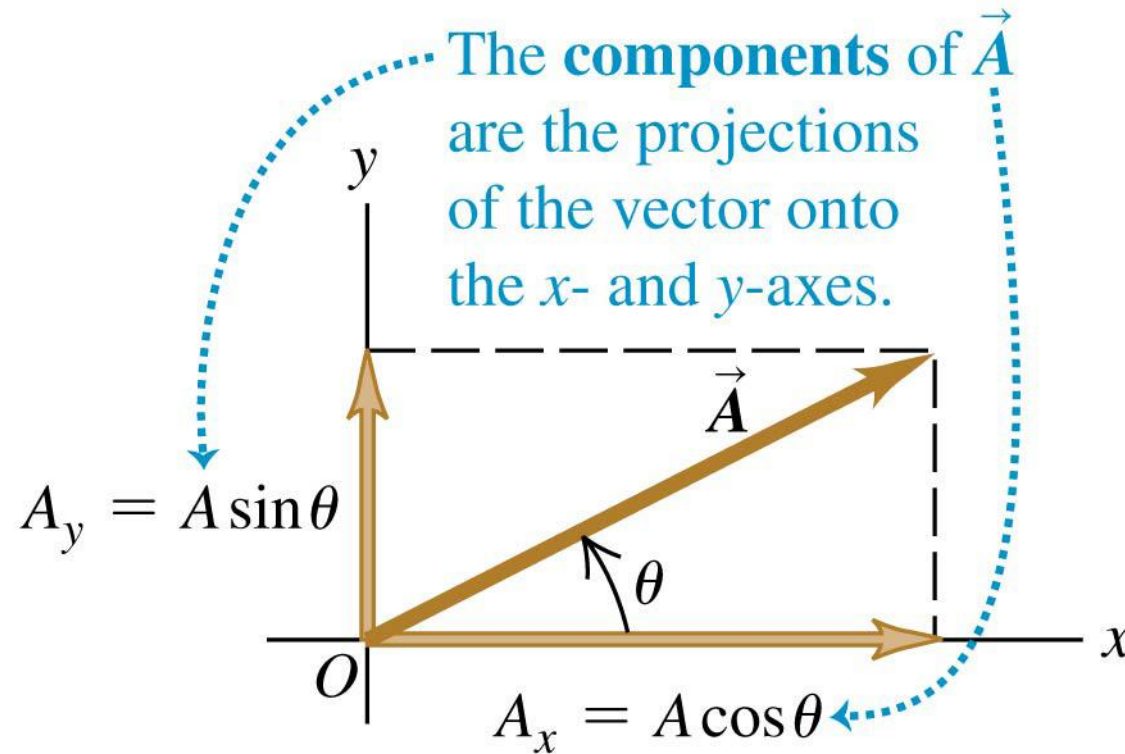
Addition of two vectors at right angles

- To add two vectors that are at right angles, first add the vectors graphically.
- Then use trigonometry to find the magnitude and direction of the sum.
- In the figure, a cross-country skier ends up 2.24 km from her starting point, in a direction of 63.4° east of north.



Components of a vector

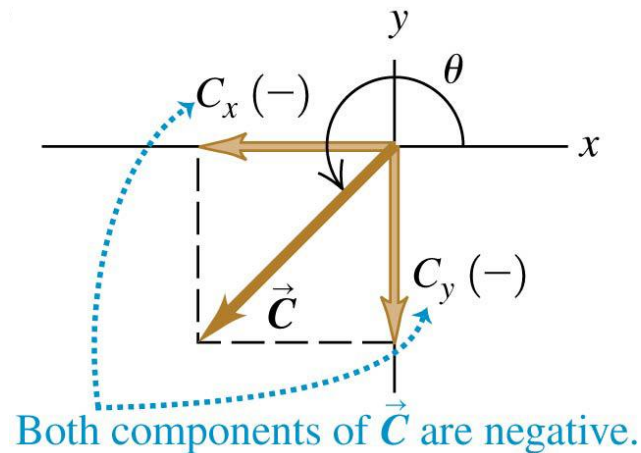
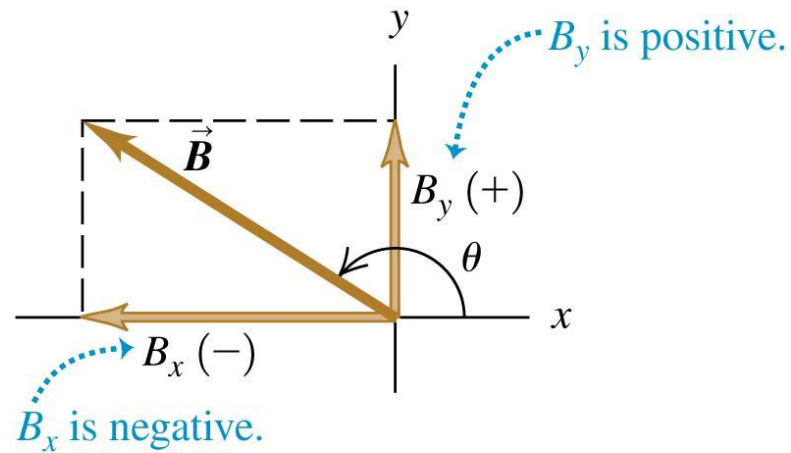
- Adding vectors graphically provides limited accuracy. Vector components provide a general method for adding vectors.
- Any vector can be represented by an x -component A_x and a y -component A_y .



In this case, both A_x and A_y are positive.

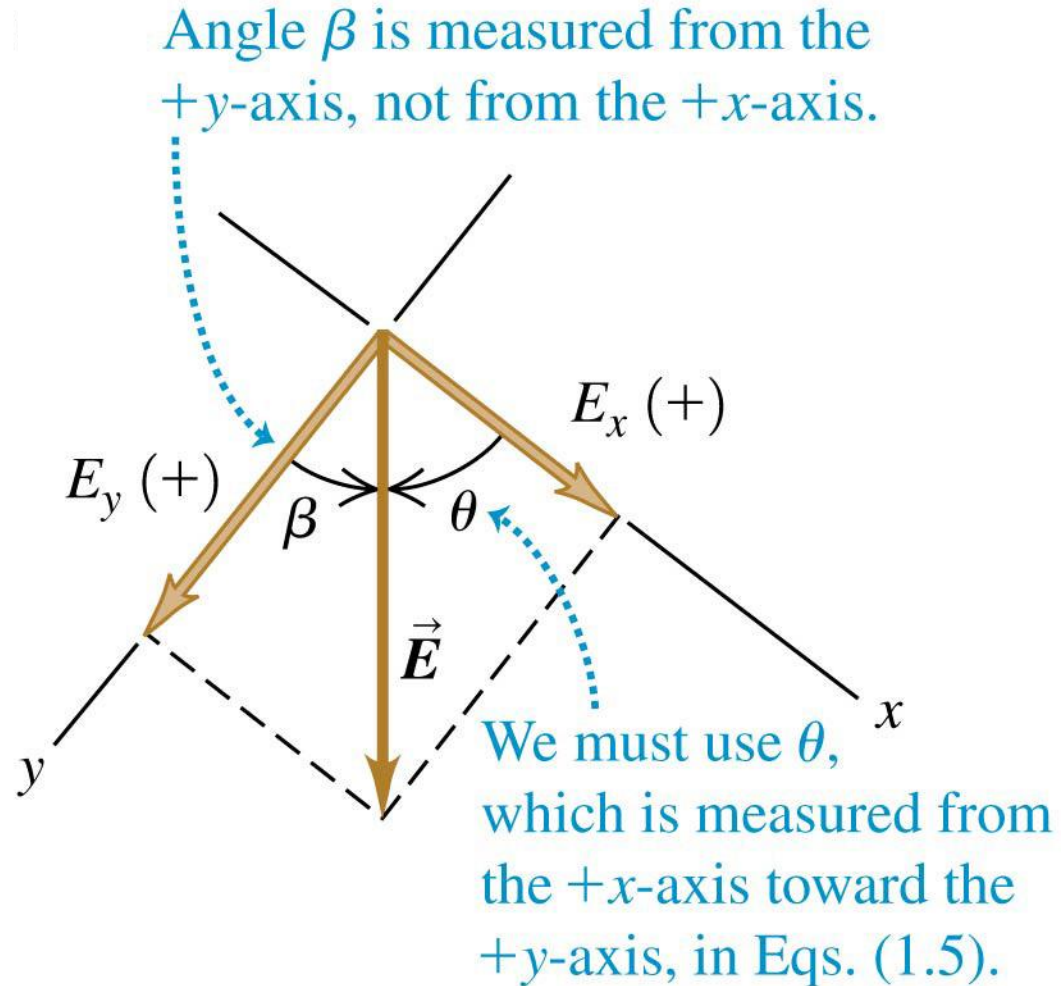
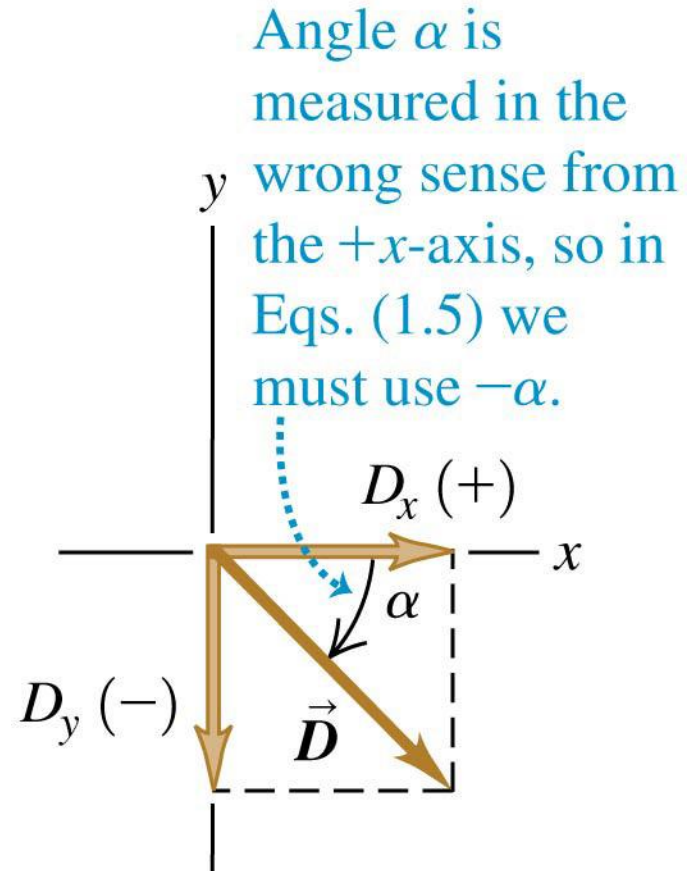
Positive and negative components

- The components of a vector may be positive or negative numbers, as shown in the figures.



Finding components

- We can calculate the components of a vector from its magnitude and direction.



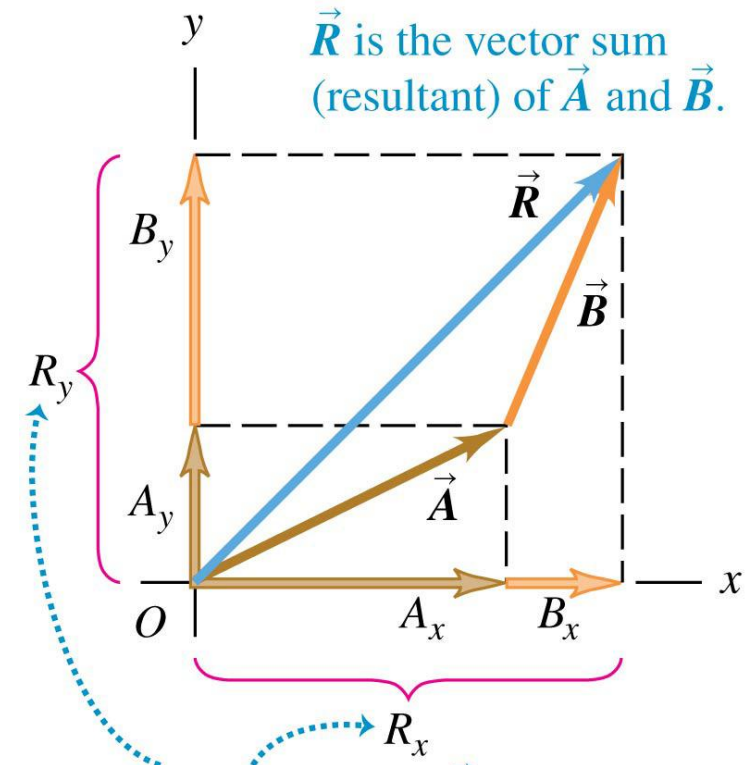
Calculations using components

- We can use the components of a vector to find its magnitude and direction:

$$A = \sqrt{A_x^2 + A_y^2} \text{ and } \tan \theta = \frac{A_y}{A_x}$$

- We can use the components of a set of vectors to find the components of their sum:

$$R_x = A_x + B_x + C_x + \dots, \quad R_y = A_y + B_y + C_y + \dots$$

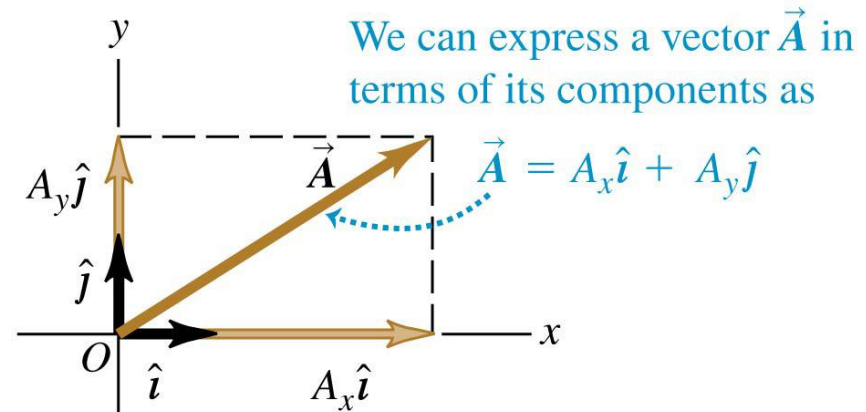
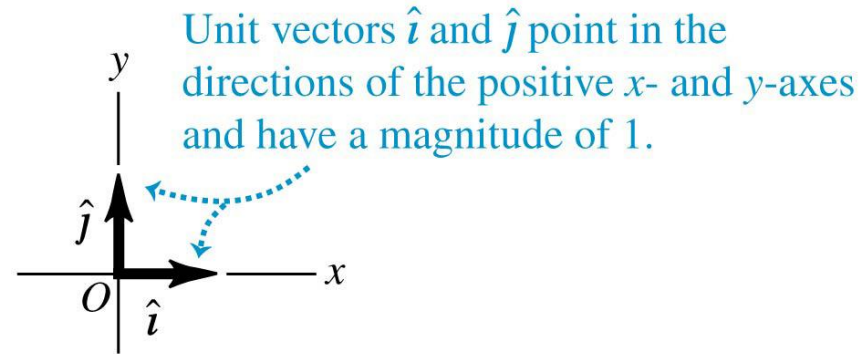


The components of $\vec{R}$ are the sums of the components of $\vec{A}$ and $\vec{B}$:

$$R_y = A_y + B_y \quad R_x = A_x + B_x$$

Unit vectors

- A **unit vector** has a magnitude of 1 with no units.
- The unit vector $\hat{i}$ points in the $+x$ -direction, $\hat{j}$ points in the $+y$ -direction, and $\hat{k}$ points in the $+z$ -direction.
- Any vector can be expressed in terms of its components as $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$.
- The unit vector can also be written as $\hat{x}, \hat{y}, \hat{z}$ or $\hat{e}_1, \hat{e}_2, \hat{e}_3$



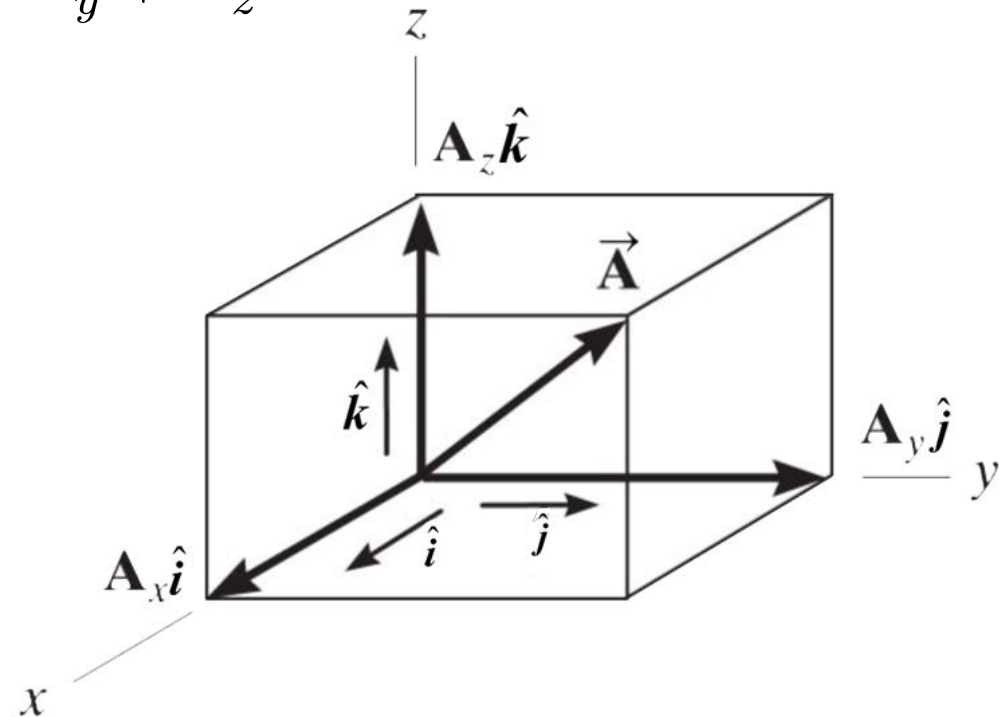
Cartesian vector notation

Vector $\mathbf{A}(\vec{A})$ is formulated by addition of its x, y, z components:

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k} = \sum_{i=1}^3 A_i \hat{e}_i$$

The *magnitude* of $\mathbf{A}$ is determined from

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$



Cartesian vector notation

The *direction* of $\mathbf{A}$ is defined in terms of its coordinate direction angles, α , β , γ , measured from the tail of $\mathbf{A}$ to the *positive* x , y , z axes. These angles are determined from the *direction cosines* which represent the $\hat{i}$, $\hat{j}$, $\hat{k}$ components of the unit vector $\vec{\mathbf{u}}_A$:

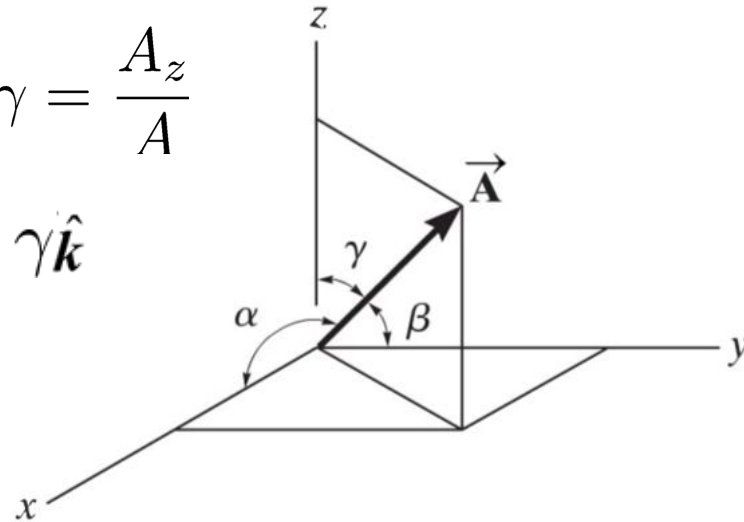
$$\vec{\mathbf{u}}_A = \frac{A_x}{A} \hat{i} + \frac{A_y}{A} \hat{j} + \frac{A_z}{A} \hat{k}$$

so that the direction cosines are

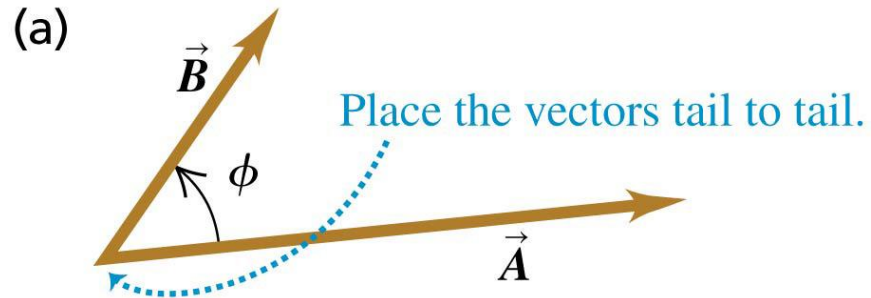
$$\cos \alpha = \frac{A_x}{A} \quad \cos \beta = \frac{A_y}{A} \quad \cos \gamma = \frac{A_z}{A}$$

Hence $\vec{\mathbf{u}}_A = \cos \alpha \hat{i} + \cos \beta \hat{j} + \cos \gamma \hat{k}$

and $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$



The scalar (dot) product

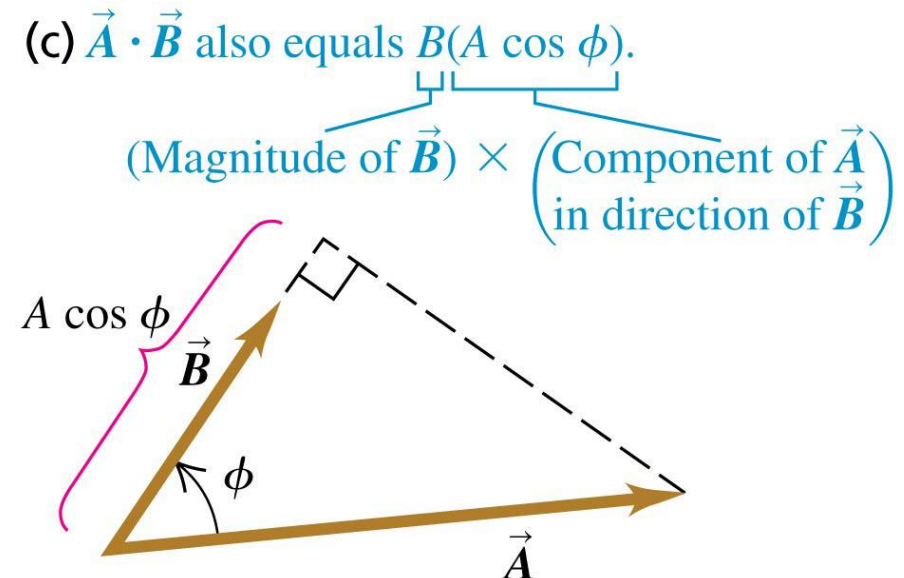
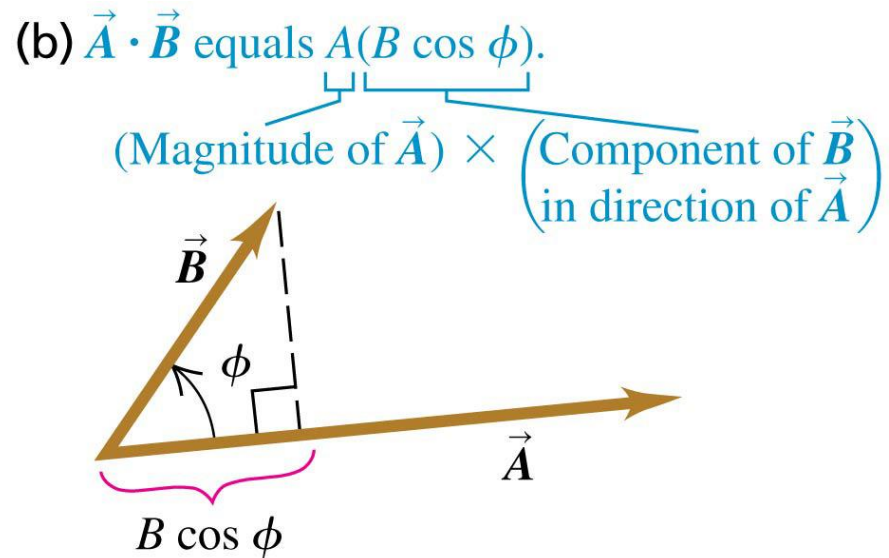


Scalar (dot) product of vectors $\vec{A}$ and $\vec{B}$

Magnitudes of $\vec{A}$ and $\vec{B}$

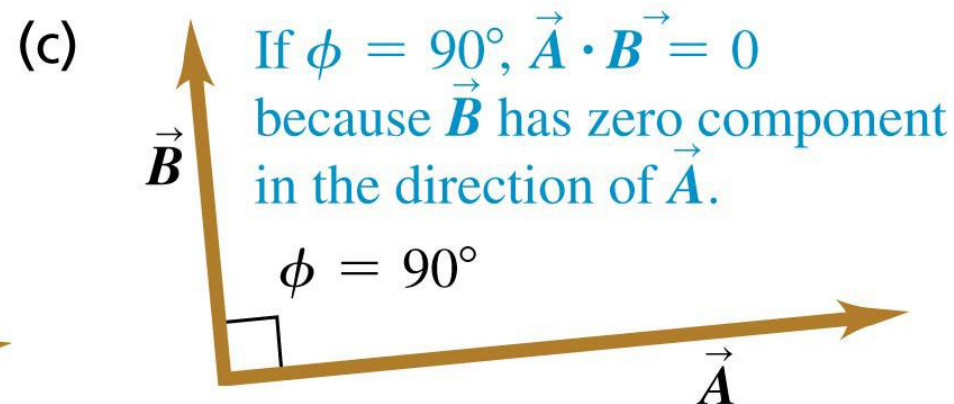
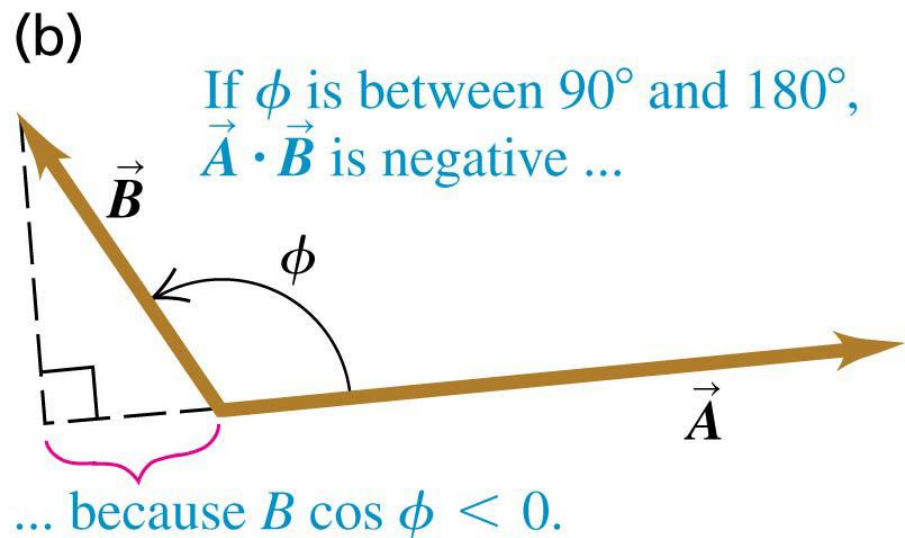
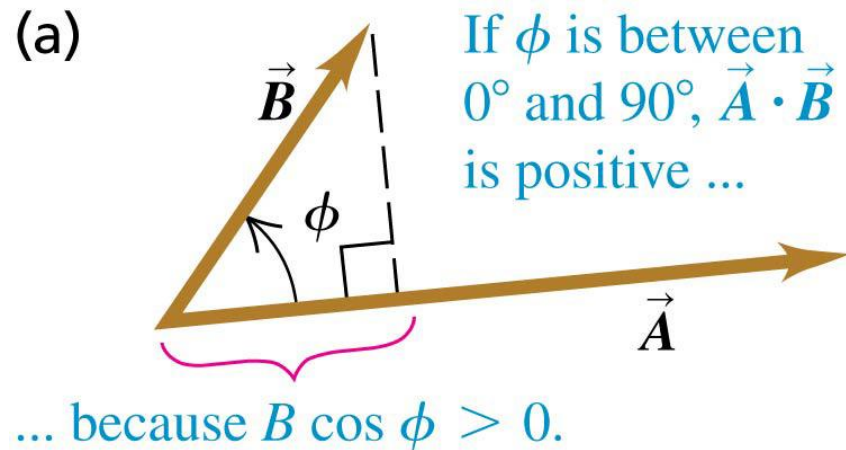
$$\vec{A} \cdot \vec{B} = AB \cos \phi = |\vec{A}| |\vec{B}| \cos \phi$$

Angle between $\vec{A}$ and $\vec{B}$ when placed tail to tail



The scalar (dot) product

The scalar product can be positive, negative, or zero, depending on the angle between $\vec{A}$ and $\vec{B}$.



The scalar (dot) product: summary

The dot product of two vectors $\vec{\mathbf{A}}$ and $\vec{\mathbf{B}}$, which yields a scalar, is defined as

$$\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} = AB \cos \theta \quad (0^\circ \leq \theta \leq 180^\circ).$$

and reads $\vec{\mathbf{A}}$ “dot” $\vec{\mathbf{B}}$. The angle θ is formed between the *tails* of $\vec{\mathbf{A}}$ and $\vec{\mathbf{B}}$

The dot product is commutative; i.e., $\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} = \vec{\mathbf{B}} \cdot \vec{\mathbf{A}}$

The distributive law is valid; i.e., $\vec{\mathbf{A}} \cdot (\vec{\mathbf{B}} + \vec{\mathbf{D}}) = \vec{\mathbf{A}} \cdot \vec{\mathbf{B}} + \vec{\mathbf{A}} \cdot \vec{\mathbf{D}}$

And scalar multiplication can be performed in any manner, i.e.,

$$m(\vec{\mathbf{A}} \cdot \vec{\mathbf{B}}) = (m\vec{\mathbf{A}}) \cdot \vec{\mathbf{B}} = \vec{\mathbf{A}} \cdot (m\vec{\mathbf{B}}) = (\vec{\mathbf{A}} \cdot \vec{\mathbf{B}})m$$

Calculating a scalar product using components

- The dot product between any two Cartesian vectors can be determined, e.g.

$$\hat{i} \cdot \hat{i} = (1)(1) \cos 0^\circ = 1 \quad \hat{i} \cdot \hat{j} = (1)(1) \cos 90^\circ = 0$$

- If $\vec{A}$ and $\vec{B}$ are expressed in Cartesian component form, then the dot product can be determined from

Scalar (dot) product
of vectors $\vec{A}$ and $\vec{B}$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Components of $\vec{A}$

Components of $\vec{B}$

- The scalar product of two vectors is the sum of the products

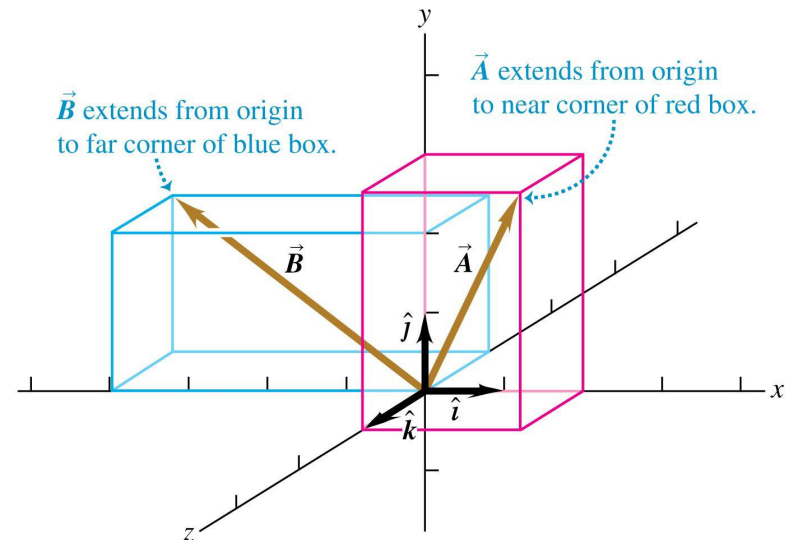
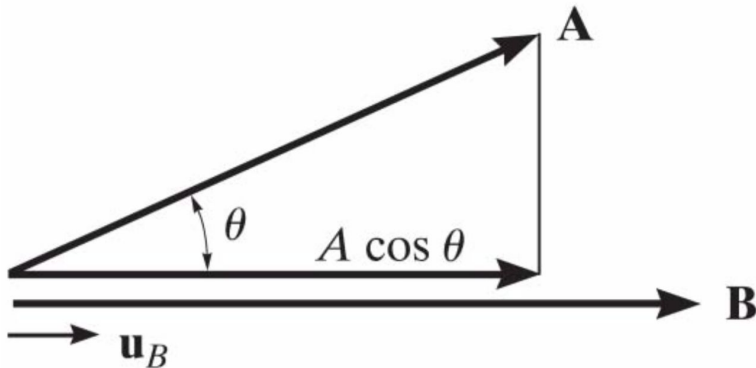
Finding an angle using the scalar product

- The dot product may be used to determine the *angle* θ formed between two vectors:

$$\theta = \cos^{-1} \left(\frac{\vec{A} \cdot \vec{B}}{AB} \right)$$

- It is also possible to find the component of a vector in a given direction using the dot product. For example, the magnitude of the component (or projection) of vector $\vec{A}$ in the direction of $\vec{B}$, is

$$A \cos \theta = \vec{A} \cdot \frac{\vec{B}}{B} = \vec{A} \cdot \vec{u}_B$$



The vector (cross) product

If the vector product (“cross product”) of two vectors is then: $\vec{C} = \vec{A} \times \vec{B}$

Magnitude of **vector (cross) product** of vectors $\vec{B}$ and $\vec{A}$

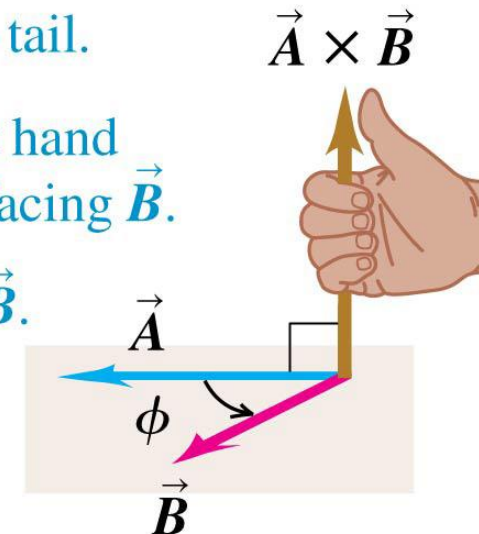
$$C = AB \sin \phi$$

Magnitudes of $\vec{A}$ and $\vec{B}$

Angle between $\vec{A}$ and $\vec{B}$ when placed tail to tail

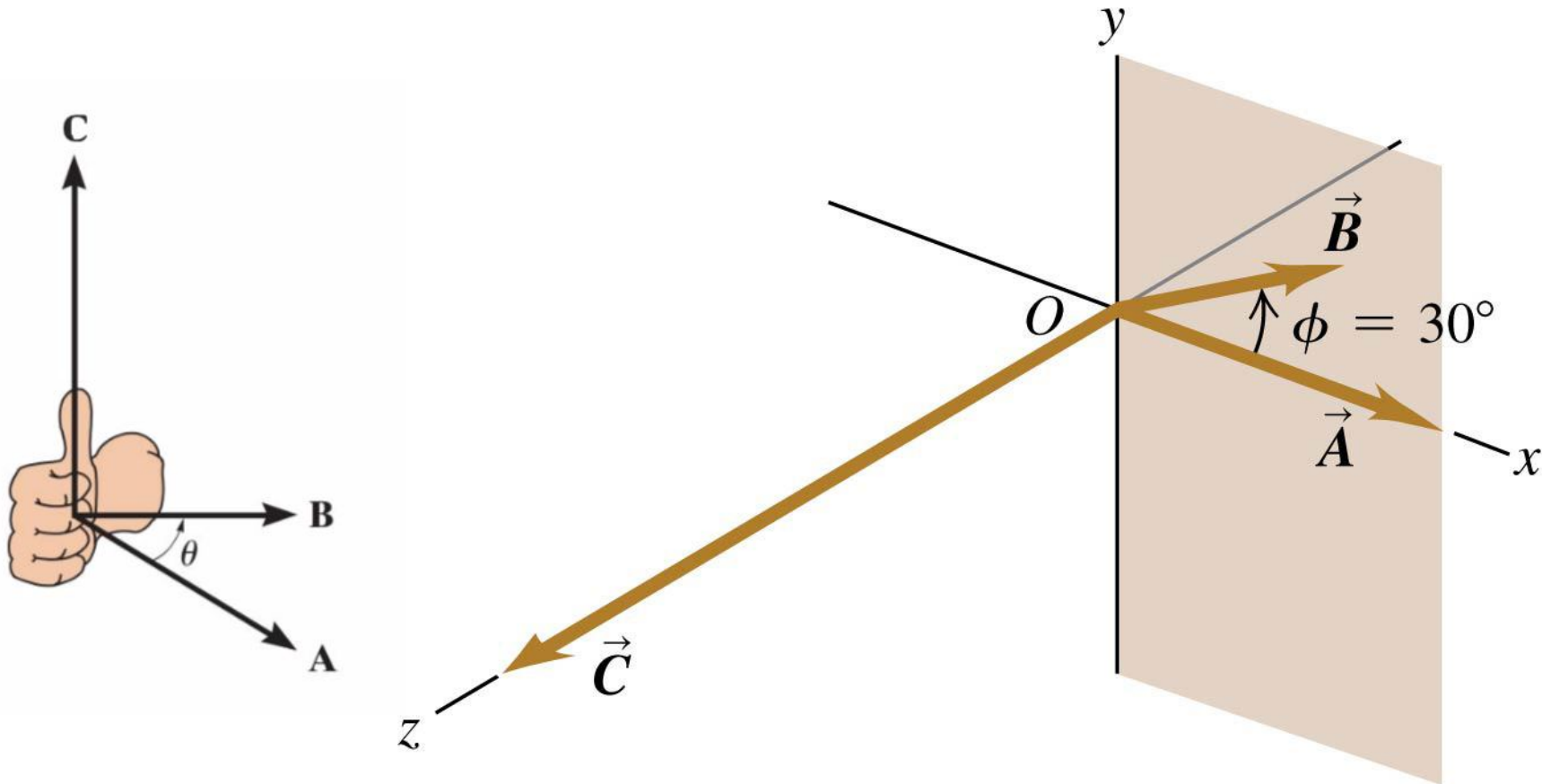
The direction of the vector product can be found using the right-hand rule:

- ① Place $\vec{A}$ and $\vec{B}$ tail to tail.
- ② Point fingers of right hand along $\vec{A}$, with palm facing $\vec{B}$.
- ③ Curl fingers toward $\vec{B}$.
- ④ Thumb points in direction of $\vec{A} \times \vec{B}$.



Calculating the vector product

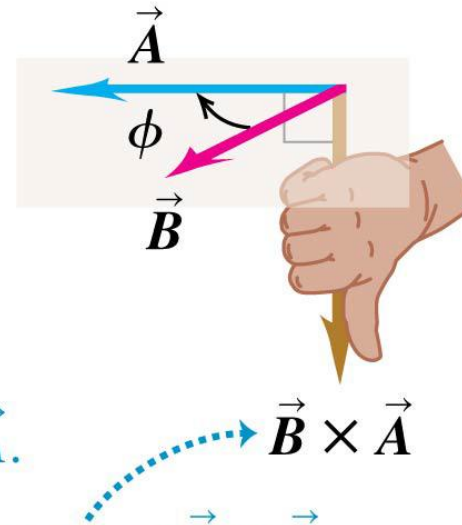
- Use $AB\sin\phi$ to find the magnitude and the right-hand rule to find the direction.



The vector (cross) product is anticommutative

$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$$

- ① Place $\vec{B}$ and $\vec{A}$ tail to tail.
- ② Point fingers of right hand along $\vec{B}$, with palm facing $\vec{A}$.
- ③ Curl fingers toward $\vec{A}$.
- ④ Thumb points in direction of $\vec{B} \times \vec{A}$.
- ⑤ $\vec{B} \times \vec{A}$ has same magnitude as $\vec{A} \times \vec{B}$ but points in opposite direction.



The vector cross product

The distributive law is valid:

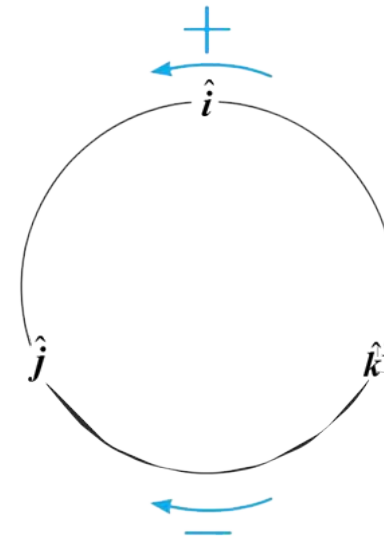
$$\vec{\mathbf{A}} \times (\vec{\mathbf{B}} + \vec{\mathbf{D}}) = \vec{\mathbf{A}} \times \vec{\mathbf{B}} + \vec{\mathbf{A}} \times \vec{\mathbf{D}}$$

And the cross product may be multiplied by a scalar m in any manner, i.e.,

$$m(\vec{\mathbf{A}} \times \vec{\mathbf{B}}) = (m\vec{\mathbf{A}}) \times \vec{\mathbf{B}} = \vec{\mathbf{A}} \times (m\vec{\mathbf{B}}) = (\vec{\mathbf{A}} \times \vec{\mathbf{B}})m$$

Cross product for any pair of Cartesian unit vectors: For example, to find $\hat{i} \times \hat{j}$, the magnitude is $(i)(j)\sin 90^\circ = 1$, and its direction $+\hat{k}$, so $\hat{i} \times \hat{j} = \hat{k}$.

A simple scheme is shown in the figure. “Crossing” two of the unit vectors in a *counterclockwise* fashion around the circle yields a *positive* third unit vector, e.g. $\hat{k} \times \hat{i} = \hat{j}$. Moving clockwise, a negative unit vector is obtained, e.g. $\hat{i} \times \hat{k} = -\hat{j}$.



The vector cross product

If $\mathbf{A}$ and $\mathbf{B}$ are expressed in Cartesian component form, then the cross product may be evaluated by expanding the determinant

$$\vec{\mathbf{C}} = \vec{\mathbf{A}} \times \vec{\mathbf{B}} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

which yields

$$\vec{\mathbf{C}} = (A_y B_z - B_y A_z) \hat{i} + (A_z B_x - B_z A_x) \hat{j} + (A_x B_y - B_x A_y) \hat{k}$$

Example

$a^n + \frac{1}{2}$
1. Show that

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{C} \times \mathbf{B}) \times \mathbf{A} = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}$$

Calculus

- We will give a brief introduction to basic calculus. You will study it in detail in math courses.
- Differentiation
- Differentiation of composite functions (chain rule)
- Integration (indefinite, definite)
- Differentiation and integration of vector functions
- Partial differentiation
- Many online resources are available. One helpful resource is:
<https://tutorial.math.lamar.edu/Classes/CalcI/CalcI.aspx>
(we used some of his examples).

Slope and the derivative

- Consider a function of x , $f(x)$
- The derivative (differentiation) of $f(x)$ with respect to x is defined as $\frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x}$
- i.e., divide Δf by Δx and gradually reduce Δx to a very small value (the meaning of $\lim_{\Delta x \rightarrow 0}$)
- Slope of AC is the derivative
- Slope of AB is $\frac{\Delta f}{\Delta x}$
- If f is distance, then the derivative is the speed

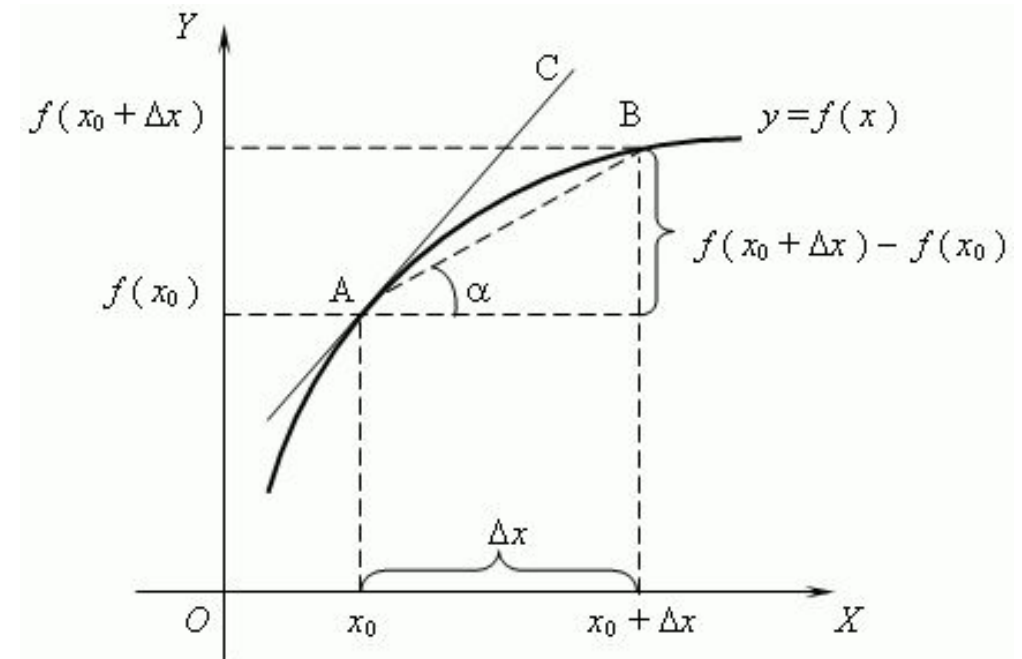


Fig. 1

Example

Given $y = x^2 - 3$ what is its derivative?

Notations

- There are many ways to denote the derivative of a function $y=f(x)$
 - $f'(x)$ the derivative of f as a function of x
 - $df(x)/dx$
 - $\frac{d}{dx}f(x)$ the differentiation operator d/dx acting on $f(x)$
 - y' y prime – the derivative of y with respect to x
 - dy/dx or $(d/dx)y$
- When the argument is time t , i.e. $y=f(t)$ we sometimes write the time derivative as a “dot” on the top, i.e.

$$\dot{y} = \frac{dy}{dt} = y'(t)$$

Notations for derivative

$$\frac{d}{dx}f$$

Leibniz

$$D_x f$$

Euler

$$f'(x)$$

Lagrange

$$\dot{f}$$

Derivative wrt time

Newton

Notations for second derivative

$$\frac{d^2}{dx^2} f \equiv \frac{d}{dx} \left(\frac{d}{dx} f \right)$$

Leibniz

$$D_x^2 f$$

Euler

$$f''(x)$$

Lagrange

$$\ddot{f}$$

Derivative wrt time

Newton

Properties of differentiation

$$(f(x) \pm g(x))' = f'(x) \pm g'(x)$$

$$(cf(x))' = cf'(x)$$

$$\text{If } f(x) = c \text{ then } f'(x) = 0$$

$$\text{If } f(x) = x^n \text{ then } f'(x) = nx^{n-1}$$

Derivative of simple functions

$$f(x) = c, \quad \frac{df}{dx} = 0 \quad ; \quad f(x) = ax, \quad \frac{df}{dx} = a$$

$$f(x) = x, \quad \frac{df}{dx} = 1 \quad ; \quad f(x) = x^2, \quad \frac{df}{dx} = 2x$$

$$f(x) = x^n, \quad \frac{df}{dx} = nx^{n-1} \quad ; \quad f(x) = \sin x, \quad \frac{df}{dx} = \cos x$$

$$f(x) = \cos x, \quad \frac{df}{dx} = -\sin x$$

$$f(x) = cy(x), \quad \frac{df}{dx} = c \frac{dy}{dx}$$

Product and quotient rule

- Product rule

If the two functions $f(x)$ and $g(x)$ are differentiable (*i.e.* the derivative exist) then the product is differentiable and,

$$(fg)' = f'g + fg'$$

- Quotient rule

If the two functions $f(x)$ and $g(x)$ are differentiable (*i.e.* the derivative exist) then the quotient is differentiable and,

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

Product rule

- The product rule can be extended to more than two functions, for instance

$$(f g h)' = f' g h + f g' h + f g h'$$

$$(f g h w)' = f' g h w + f g' h w + f g h' w + f g h w'$$

- Proof:

Composite functions and the chain rule

- If we define

$$f(z) = \sqrt{z} \quad g(z) = 5z - 8$$

- Then we can write a composite function

$$R(z) = (f \circ g)(z) = f(g(z)) = \sqrt{5z - 8}$$

- We can use the chain rule to differentiate a composite function

Suppose that we have two functions $f(x)$ and $g(x)$ and they are both differentiable.

1. If we define $F(x) = (f \circ g)(x)$ then the derivative of $F(x)$ is,

$$F'(x) = f'(g(x)) g'(x)$$

2. If we have $y = f(u)$ and $u = g(x)$ then the derivative of y is,

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

Example 1

- Differentiate $R(z) = \sqrt{5z - 8}$.

Summary

In general:

$$F'(x) = \underbrace{f'}_{\text{derivative of outside function}} \underbrace{(g(x))}_{\text{inside function left alone}} \underbrace{g'(x)}_{\text{times derivative of inside function}}$$

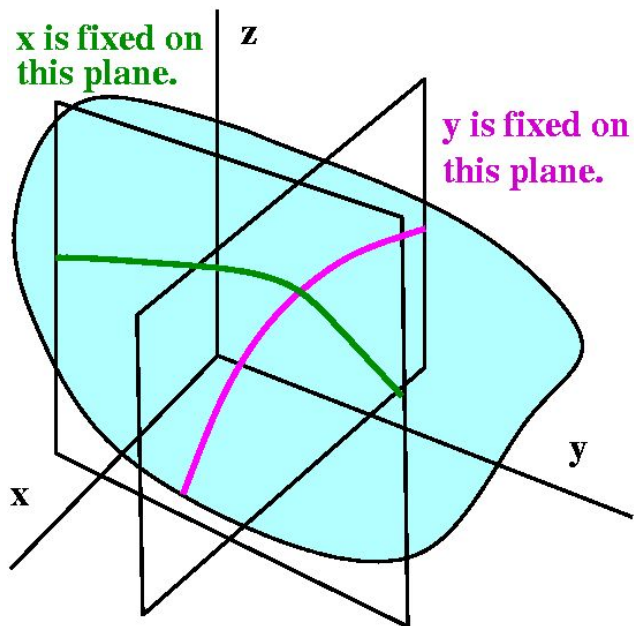
Example 2

If the path in polar coordinates is given as $r = 5\theta^2$, where θ is a known function of time, find $\dot{r}$.

.

Partial differentiation (partial derivative)

- The function is a function of two variables, x, y $z = f(x, y)$
- Keep one variable x or y constant and differentiate with respect to the other variable y or x .



$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial x} = \frac{df}{dx} \text{ assuming } y \text{ a constant}$$

$$\frac{\partial z}{\partial y} = \frac{\partial f}{\partial y} = \frac{df}{dy} \text{ assuming } x \text{ a constant}$$

Newton approximation

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} \quad \longrightarrow \quad f(x) \approx f(0) + f'(0)x$$

if $|x| \ll 1$

$$e^x \approx 1 + x$$

$$\sin x \approx x$$

$$\cos x \approx 1 - \frac{x^2}{2}$$

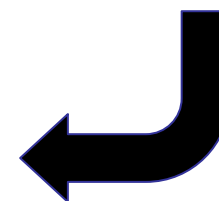
$$\tan x \approx x$$

$$\ln(1 - x) \approx -x$$

$$\ln(1 + x) \approx x$$

$$(1 + x)^\alpha \approx 1 + \alpha x$$

$$(1 - x)^\alpha \approx 1 - \alpha x$$



Indefinite integral

- Given a function, $f(x)$, an anti-derivative of $f(x)$ is any function $F(x)$ such that $F'(x) = f(x)$.
- If $F(x)$ is any anti-derivative of $f(x)$ then the most general anti-derivative of $f(x)$ is called an **indefinite integral** and denoted
$$\int f(x)dx = F(x) + c$$
- $\int$ is called the **integral symbol**, $f(x)$ is called the **integrand**, x is called the **integration variable** and “c” the **constant of integration**.
- The process of finding the indefinite integral is called **integration** or **integrating $f(x)$** . If we need to be specific about the integration variable we say that we are **integrating $f(x)$ with respect to x** .

Indefinite integral

- Example: Which function has the following derivative?

$$f'(x) = x^4 + 3x - 9$$

Properties of indefinite integral

- The integration is linear, i.e.

$$\int f(x) \pm g(x) dx = \int f(x) dx \pm \int g(x) dx$$

$$\int k f(x) dx = k \int f(x) dx$$

k is any number.

Indefinite integral of common functions

$$\int a dx = ax + c$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c$$

$$\int \cos x dx = \sin x + c$$

$$\int \sin x dx = -\cos x + c$$

Differentiation and integration of Vectors

- The rules for differentiation and integration of the sums and products of scalar functions also apply to vector functions. Consider, for example, the two vector functions $\mathbf{A}(s)$ and $\mathbf{B}(s)$. Provided these functions are smooth and continuous for all s , then

$$\frac{d}{ds}(\mathbf{A} + \mathbf{B}) = \frac{d\mathbf{A}}{ds} + \frac{d\mathbf{B}}{ds}$$

$$\int (\mathbf{A} + \mathbf{B})ds = \int \mathbf{A}ds + \int \mathbf{B}ds$$

$$\frac{d}{ds}(\mathbf{A} \cdot \mathbf{B}) = \frac{d\mathbf{A}}{ds} \cdot \mathbf{B} + \mathbf{A} \cdot \frac{d\mathbf{B}}{ds}$$

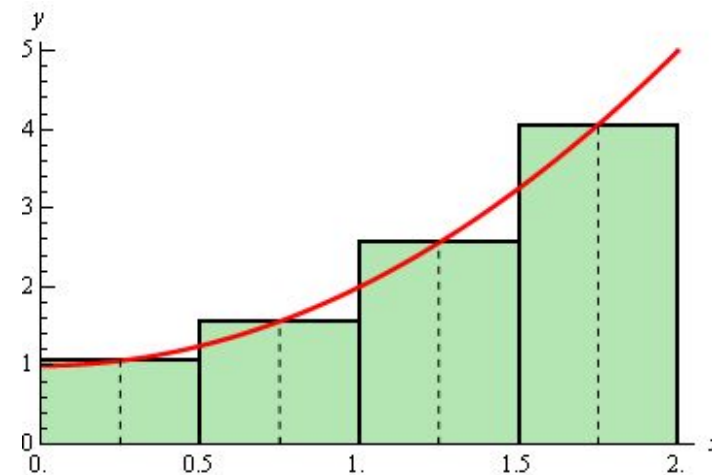
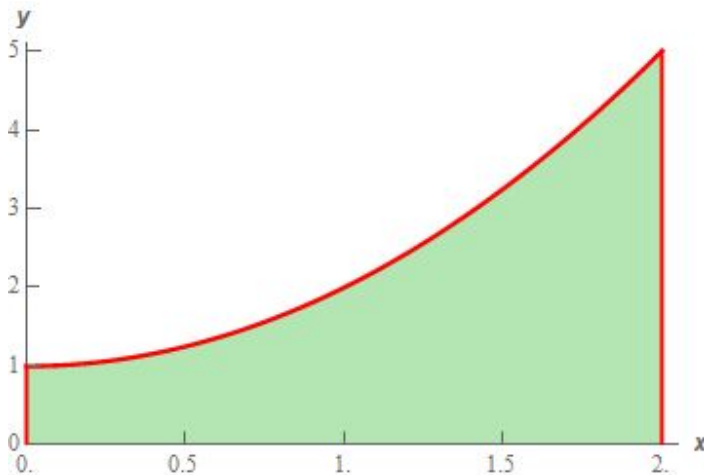
$$\frac{d}{ds}(\mathbf{A} \times \mathbf{B}) = \frac{d\mathbf{A}}{ds} \times \mathbf{B} + \mathbf{A} \times \frac{d\mathbf{B}}{ds}$$

Area problem

- Suppose we want to determine the area under $f(x) = x^2 + 1$ on $[0,2]$ as shown in the figure.
- We can estimate the area by dividing up the interval into n subintervals, each of width $\Delta x = (b - a)/n$. So the area can be estimated as

$$A \approx \sum_{i=1}^n f(x_i^*) \Delta x$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$



Definite integral

- Given a function $f(x)$ that is continuous on the interval $[a, b]$ we divide the interval into n subintervals of equal width, Δx , and from each interval choose a point, x_i^* , then the **definite integral** of $f(x)$ from a to b is

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

a, b can be $\pm\infty$

- Method to calculate: take the indefinite integral of $f(x)$, evaluate at b and a , and take the difference. Example:

$$\int_0^2 x^2 dx = \left. \frac{x^3}{3} \right|_0^2 = \frac{2^3 - 0^3}{3} = \frac{8}{3}$$

(note that the constant c makes no contribution)

Properties of definite integral

- Interchanging the limits gets you a minus sign:

$$\int_a^b f(x) \, dx = - \int_b^a f(x) \, dx$$

- If the upper and lower limits are the same then the area will be 0

$$\int_a^a f(x) \, dx = 0$$

- For any number c ,

$$\int_a^b c f(x) \, dx = c \int_a^b f(x) \, dx$$

- Definite integrals can be broken up across a sum or difference

$$\int_a^b f(x) \pm g(x) \, dx = \int_a^b f(x) \, dx \pm \int_a^b g(x) \, dx$$

Properties of definite integral

- (Note that c does not have to be between a and b)

$$\int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx$$

- Changing the integration variable will not affect the answer

$$\int_a^b f(x) \, dx = \int_a^b f(t) \, dt$$

- For any number c ,

$$\int_a^b c \, dx = c(b - a)$$

Fundamental theorem of calculus

- Suppose $f(x)$ is a continuous function on $[a, b]$ and also suppose that $F(x)$ is any anti-derivative for $f(x)$. Then,

$$\int_a^b f(x) dx = F(x)|_a^b = F(b) - F(a)$$

Example

$$\int_1^2 y^2 + y^{-2} dy = ?$$

Useful mathematical expressions

Quadratic Formula

$$\text{If } ax^2 + bx + c = 0, \text{ then } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Hyperbolic Functions

$$\sinh x = \frac{e^x - e^{-x}}{2}, \cosh x = \frac{e^x + e^{-x}}{2}, \tanh x = \frac{\sinh x}{\cosh x}$$

Trigonometric Identities

$$\sin \theta = \frac{A}{C}, \csc \theta = \frac{C}{A}$$

$$\cos \theta = \frac{B}{C}, \sec \theta = \frac{C}{B}$$

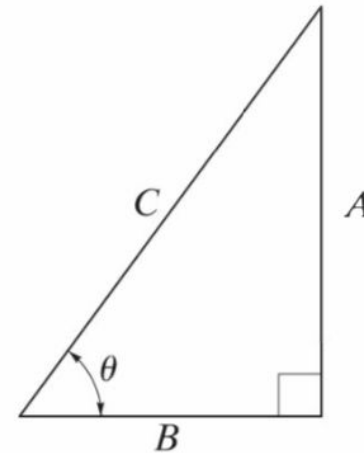
$$\tan \theta = \frac{A}{B}, \cot \theta = \frac{B}{A}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin(\theta \pm \phi) = \sin \theta \cos \phi \pm \cos \theta \sin \phi$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos(\theta \pm \phi) = \cos \theta \cos \phi \mp \sin \theta \sin \phi$$



Useful mathematical expressions

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\cos \theta = \pm \sqrt{\frac{1 + \cos 2\theta}{2}}, \sin \theta = \pm \sqrt{\frac{1 - \cos 2\theta}{2}}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$1 + \tan^2 \theta = \sec^2 \theta \quad 1 + \cot^2 \theta = \csc^2 \theta$$

Power-Series Expansions

$$\sin x = x - \frac{x^3}{3!} + \dots \quad \sinh x = x + \frac{x^3}{3!} + \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \dots \quad \cosh x = 1 + \frac{x^2}{2!} + \dots$$

Derivatives

$$\frac{d}{dx}(u^n) = nu^{n-1} \frac{du}{dx}$$

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$\frac{d}{dx}(\cot u) = -\csc^2 u \frac{du}{dx}$$

Useful mathematical expressions

$$\frac{d}{dx}(\sec u) = \tan u \sec u \frac{du}{dx}$$

Integrals

$$\frac{d}{dx}(\csc u) = -\csc u \cot u \frac{du}{dx}$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$$

$$\frac{d}{dx}(\sin u) = \cos u \frac{du}{dx}$$

$$\int \frac{dx}{a+bx} = \frac{1}{b} \ln(a+bx) + C$$

$$\frac{d}{dx}(\cos u) = -\sin u \frac{du}{dx}$$

$$\int \frac{dx}{a+bx^2} = \frac{1}{2\sqrt{-ba}} \ln \left[\frac{a+x\sqrt{-ab}}{a-x\sqrt{-ab}} \right] + C, ab < 0$$

$$\frac{d}{dx}(\tan u) = \sec^2 u \frac{du}{dx}$$

$$\int \frac{x dx}{a+bx^2} = \frac{1}{2b} \ln(bx^2+a) + C,$$

$$\frac{d}{dx}(\sinh u) = \cosh u \frac{du}{dx}$$

$$\int \frac{x^2 dx}{a+bx^2} = \frac{x}{b} - \frac{a}{b\sqrt{ab}} \tan^{-1} \frac{x\sqrt{ab}}{a} + C, ab > 0$$

$$\frac{d}{dx}(\cosh u) = \sinh u \frac{du}{dx}$$

$$\int \frac{dx}{a^2-x^2} = \frac{1}{2a} \ln \left[\frac{a+x}{a-x} \right] + C, a^2 > x^2$$

Useful mathematical expressions

$$\int \sqrt{a + bx} \, dx = \frac{2}{3b} \sqrt{(a + bx)^3} + C$$

$$\int x \sqrt{a + bx} \, dx = \frac{-2(2a - 3bx) \sqrt{(a + bx)^3}}{15b^2} + C$$

$$\int x^2 \sqrt{a + bx} \, dx = \frac{2(8a^2 - 12abx + 15b^2x^2) \sqrt{(a + bx)^3}}{105b^3} + C$$

$$\int \sqrt{a^2 - x^2} \, dx = \frac{1}{2} \left[x \sqrt{a^2 - x^2} + a^2 \sin^{-1} \frac{x}{a} \right] + C, a > 0$$

$$\int x \sqrt{x^2 \pm a^2} \, dx = \frac{1}{3} \sqrt{(x^2 \pm a^2)^3} + C$$

$$\begin{aligned} \int x^2 \sqrt{a^2 - x^2} \, dx &= -\frac{x}{4} \sqrt{(a^2 - x^2)^3} \\ &\quad + \frac{a^2}{8} \left(x \sqrt{a^2 - x^2} + a^2 \sin^{-1} \frac{x}{a} \right) + C, a > 0 \end{aligned}$$

$$\int \sqrt{x^2 \pm a^2} \, dx = \frac{1}{2} \left[x \sqrt{x^2 \pm a^2} \pm a^2 \ln(x + \sqrt{x^2 \pm a^2}) \right] + C$$

Useful mathematical expressions

$$\int x\sqrt{a^2 - x^2} dx = -\frac{1}{3}\sqrt{(a^2 - x^2)^3} + C$$

$$\int x^2\sqrt{x^2 \pm a^2} dx = \frac{x}{4}\sqrt{(x^2 \pm a^2)^3} \mp \frac{a^2}{8}x\sqrt{x^2 \pm a^2} \\ - \frac{a^4}{8}\ln(x + \sqrt{x^2 \pm a^2}) + C$$

$$\int \frac{dx}{\sqrt{a + bx}} = \frac{2\sqrt{a + bx}}{b} + C$$

$$\int \frac{x dx}{\sqrt{x^2 \pm a^2}} = \sqrt{x^2 \pm a^2} + C$$

$$\int \frac{dx}{\sqrt{a + bx + cx^2}} = \frac{1}{\sqrt{c}}\ln\left[\sqrt{a + bx + cx^2} \right. \\ \left. + x\sqrt{c} + \frac{b}{2\sqrt{c}}\right] + C, c > 0$$

$$= \frac{1}{\sqrt{-c}}\sin^{-1}\left(\frac{-2cx - b}{\sqrt{b^2 - 4ac}}\right) + C, c < 0$$

Useful mathematical expressions

$$\int \sin x \, dx = -\cos x + C$$

$$\int \cos x \, dx = \sin x + C$$

$$\int x \cos(ax) \, dx = \frac{1}{a^2} \cos(ax) + \frac{x}{a} \sin(ax) + C$$

$$\int x^2 \cos(ax) \, dx = \frac{2x}{a^2} \cos(ax) + \frac{a^2 x^2 - 2}{a^3} \sin(ax) + C$$

$$\int e^{ax} \, dx = \frac{1}{a} e^{ax} + C$$

$$\int x e^{ax} \, dx = \frac{e^{ax}}{a^2} (ax - 1) + C$$

$$\int \sinh x \, dx = \cosh x + C$$

$$\int \cosh x \, dx = \sinh x + C$$

Software

- Wolfram Alpha

<https://www.wolframalpha.com/>



In[25]:= **Integrate**[$x \text{ Exp}[-x]$, { x , 0, 1}]

Out[25]= $\frac{-2 + e}{e}$

In[23]:= **Integrate**[$x \text{ Exp}[-x^2]$, { x , 0, 1}]

Out[23]= $\frac{-1 + e}{2 e}$

In[24]:= **Integrate**[$\text{Exp}[-x^2]$, { x , -Infinity, +Infinity}]

Out[24]= $\sqrt{\pi}$